

~\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\A7899B7813768C94399D6D5C16872094B93DC689\transfers\2026-24\Student_Data_Manager.py

```
1 # Mini project-1
2
3 # Student Data Manager
4 # Build a Student Record System Using OOP
5 # Features: Add, Update, Delete, Search Records
6
7 # CLASS STUDENT
8 class Student:
9     def __init__(self, roll_no, name, age, branch):
10         self.roll_no = roll_no
11         self.name = name
12         self.age = age
13         self.branch = branch
14
15
16     def display(self):
17         print("Roll No:", self.roll_no)
18         print("Name", self.name)
19         print("Age", self.age)
20         print("Branch:", self.branch)
21
22 students = []
23
24 # ADD STUDENT
25 def add_student():
26     roll_no = int(input("Enter roll no: "))
27
28
29 # DUPLICATE ROLL NUMBER CHECK
30 for student in students:
31     if student.roll_no == roll_no:
32         print("Roll Number Already Exists!")
33         return
34
35
36 name = input("Enter name: ")
37 age = int(input("Enter age: "))
38 branch = input("Enter branch: ")
39
40 student = Student(roll_no, name, age, branch)
41 students.append(student)
42
43 print("Student Added Successfully!")
44
45
46 # SEARCH STUDENT BY ROLL NUMBER:
47 def search_student():
48     roll = int(input("Enter roll number to search: "))
49
50     for student in students:
51         if student.roll_no == roll:
52             print("Student Found!")
53             student.display()
54             return
55
56     print("Student Not Found!")
57
58
59 # UPDATE STUDENT
60 def update_student():
61     roll = int(input("Enter roll number to update: "))
62
63     for student in students:
64         if student.roll_no == roll:
65             student.name = input("Enter New Name: ")
66             student.age = int(input("Enter New Age: "))
67             student.branch = input("Enter New Branch: ")
```

```

68
69         print("Student Updated Successfully!")
70         return
71
72     print("Student Not Found!")
73
74 # DELETE STUDENT
75 def delete_student():
76     roll = int(input("Enter roll no to delete: "))
77
78     for student in students:
79         if student.roll_no == roll:
80             students.remove(student)
81             print("Student Deleted Successfully!")
82             return
83
84     print("Student Not Found!")
85
86
87
88 def display_students():
89     if len(students) == 0:
90         print("No Students Found!")
91         return
92
93     for student in students:
94         student.display()
95         print()
96
97
98 # MAKE A CHOICE
99 while True:
100     print("\n==== STUDENT DATA MANAGER ====")
101     print("1.Add Student")
102     print("2.Display Students")
103     print("3.Search Student")
104     print("4.Update Student")
105     print("5.Delete Student")
106     print("6.Exit")
107
108     choice = int(input("Enter your choice: "))
109
110     if choice == 1:
111         num = int(input("How many students do you want to add? "))
112
113         for i in range(num):
114             print(f"\nEnter details of student {i+1}")
115             add_student()
116
117     elif choice == 2:
118         display_students()
119
120     elif choice == 3:
121         search_student()
122
123     elif choice == 4:
124         update_student()
125
126     elif choice == 5:
127         delete_student()
128
129     elif choice == 6:
130         print("Thank You!")
131         break
132
133     else:
134         print("Invalid Choice!")
135

```